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Recreational Cannabis Legalization and Illicit Drugs: Drug Usage, Mortality, and Darknet Transactions

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Abstract
Recreational cannabis legalization has become an unprecedented trend in the United States, arousing heated debates on its implications for public health. Legalizing access to recreational cannabis may shrink the illicit market as it could be a substitute for illicit substances—the substitution effect. Alternatively, recreational cannabis may serve as a gateway drug to illicit drugs and raise the risk of using illicit substances—the gateway effect. Using novel darknet transactional data along with public health databases, we add evidence to the debate by empirically investigating the impact of recreational cannabis legalization on illicit drug use and mortality. Capitalizing on a quasi-experiment opportunity in which U.S. states legalized recreational cannabis at different times, we found a significant increase in the mortality of illicit drug overdose, along with the initiation and use of illicit drugs after the policy, i.e., the gateway effect. This effect is evident in a significant surge of Bitcoin transactions on darknets—a major source of illicit drugs—after the policy. It is also particularly salient in states allowing higher dosage possession and lower excise taxes on recreational cannabis. Our study contributes novel data to address the public health debate and provide policymakers with timely regulatory recommendations on recreational cannabis legalization.

Keywords
Recreational Cannabis Legalization, Illicit Drugs, Darknet, Overdose Mortality, Public Health

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1 Introduction

In recent years, the United States has witnessed a sweeping trend of Recreational Cannabis Legalization (RCL or the policy hereafter). RCL refers to the public policy that allows the retail sale of cannabis for recreational use by adults (Hall, 2015). Since the state of Washington first legalized recreational cannabis in 2012, the policy has taken effect in 19 states as of July 2022, and more states are anticipated to join the list.

The passing of RCL has never been without heated debates. Advocates argue that the policy combats the illicit economy (Hsu and Kovács, 2021; Livingston et al., 2017), increases tax revenue (Felson et al., 2019; Leyton, 2019), and reduces crime rates (Hall and Lynskey, 2016). Specific to its impact on public health, advocates believe that the policy promotes recreational cannabis as a substitute for illicit substances such as heroin and cocaine, which can cause a higher risk of overdose mortality (Chan et al., 2020; Janik et al., 2017). As marijuana becomes easier to access, users consume fewer illicit drugs[1] and experience lower overdose mortality. We name this *the substitution effect*. Critics, in contrast, are concerned that the policy may provide a gateway that increases the likelihood of users engaging in the subsequent use of illicit and more harmful illicit substances, causing increased overdose mortality (Resko et al., 2019) and worsening drug abuse problems (Park et al., 2020). We name this *the gateway effect*.

Although both the substitution effect and the gateway effect are possible, it is *ex-ante* unclear which effect dominates the policy's impact due to the inconsistency of evidence as well as the lack of data supporting the underlying reasonings behind the impact. Would the overdose mortality of illicit drugs increase (or decrease) after states legalize recreational cannabis? Is the increasing availability and acceptability of

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recreational cannabis at the cost of the drug overdose epidemic? These are questions without clear answers in the literature yet crucial to the political proposition of RCL. As the recreational cannabis markets in the United States increasingly proliferate, studies on the public health consequences of the policy are warranted, as well as public and professional education about the risk-benefit tradeoff. If the paradigm of recreational cannabis involves a shift toward a more liberal regulatory stance, it is critical to understand its impact on illicit drug use and what policy intervention may protect public health while improving social welfare.

Whether recreational cannabis provides a substitute or a gateway to illicit drugs is an empirical question and requires data-driven evidence and reasoning. In this paper, we empirically investigate the impact of legalizing recreational cannabis by leveraging a unique opportunity of several U.S. states that legalized recreational cannabis at different times. Our empirical strategy examines the policy impacts on various outcomes of illicit drug use to determine the policy's effect. To see this, the two distinctive hypotheses, the substitution effect and the gateway effect, lead to different predictions about the policy's impacts. If the substitution effect dominates, the policy, by allowing mass access to recreational cannabis, will encourage users to consume recreational cannabis as a substitute for illicit drugs. As a result, the policy will likely result in a decrease in the initiation and use of illicit drugs and the associated overdose mortality. Conversely, if the gateway effect dominates, the policy will stimulate the desire to initiate and use illicit drugs. Consequently, the overdose mortality of illicit drugs will likely increase. In other words, the policy's impacts (on overdose mortality of illicit drugs and other outcomes) would serve as evidence of whether recreational cannabis is a substitution or gateway drug.

We further dive into the reasonings behind the policy's impacts by investigating (a) whether recreational cannabis use indeed increased after the policy (a premise for either the substitution effect or the gateway effect), (b) whether the varying levels of leniency in states passing the policy affect users' desire to consume illicit drugs, and (c) was there a change of transactions on darknets, a popular and major source of illicit drugs, after the policy. These three sets of evidence will provide empirical evidence that can speak to how the policy affects the overdose mortality of illicit drugs. Finally, we examine how the policy's impacts vary across states to generate localized policymaking implications.

Our main specification is a standard difference-in-differences (DID) approach using state-by-year-month panel data. In our DID design, our treated group consists of eleven U.S. states that passed RCL between January 2012 and December 2018 (i.e., the policy-affected states). We construct a comparison group of U.S. states that can best mirror those policy-affected states but have not experienced the policy (i.e., the policy-unaffected states). We deploy alternative specifications such as a relative time model, DID with multiple time periods (CSDID), and synthetic DID (SDID) (Angrist and Pischke, 2008; Arkhangelsky et al., 2021; Callaway and Sant'Anna, 2021) to ensure the robustness of our results while ruling out alternative explanations for the impact. For easy reference, we furnish a plethora of these tests supporting the main results, as shown in Table A1 of the Web Appendix.

Our novel data are collected from multiple sources. First, we collected data on overdose mortality of illicit drugs and their counterparts (e.g., legal drugs[2]) from the Centers for Disease Control and Prevention (CDC). The data contain monthly overdose mortality records of 50 states and the District of Columbia in the United States (for simplicity, 51 states hereafter). For each state, we obtained data on the initiation and use of illicit drugs from the annual National Survey on Drug Use and Health (NSDUH). These two datasets allowed us to measure how the policy affected the use of illicit drugs in comparison with other non-illicit substances, thereby revealing whether the primary effect of the policy is substitution or gateway. Second, we collected Bitcoin transactions on darknet markets from WalletExplorer.com, a platform that monitors Bitcoin wallets related to darknet markets. For each Bitcoin transaction, we traced darknet market buyers' IP addresses using BlockCypher.com and inferred their geolocations using IPInfo.io. This novel data allows us to examine the change in illicit drug transactions in darknet markets after the policy, thus adding evidence to the reasoning behind the policy's impacts. Third, we provided additional evidence on the reasoning by analyzing the leniency of the RCL policy. If, with a more lenient RCL policy, reflected in higher dosage allowance and a lower excise tax rate, we observe an increase in overdose mortality from illicit drugs, the policy is more likely to have a gateway effect. Finally, as the RCL impact is not immune to local characteristics, we obtained data on the economic and demographic characteristics from the U.S. Census Bureau's annual American Community Surveys (ACS). The final sample, from combining multiple datasets, consists of 51 U.S. states and spans a period from January 2012 to December 2018—a total of 84 months, covering a sufficiently large time window for the policy impacts to manifest.

Our empirical findings are mainly four-fold. The *first* set of findings concerns the policy impacts on multiple outcomes of illicit drug use (to determine the effect of RCL). We find that the overdose mortality caused by illicit drugs surged by 34.6% after the policy, consistent with the prediction under the gateway effect. Similar results are also shown in the relative time model, indicating the policy indeed escalated illicit drug use. We continue to find that, different from the surge in the overdose mortality of illicit drugs, the use and overdose mortality of legal drugs and alcohol remain unchanged after the policy. It is thus likely the increased overdose mortality of illicit drugs is a result of the gateway effect rather than the overall trend of increased addictive substance use.

If the policy's gateway effect holds true, we shall see more users start using illicit drugs after the policy. Our *second* set of findings provides direct evidence to support this assumption. Our results first show that the policy, by providing access to recreational cannabis, significantly increased the number of people who initiated and used recreational cannabis by 20.7% and 9.1%, respectively, confirming the premise of the policy's impact. More importantly, we find that the policy significantly increased the number of people who initiated and used illicit drugs to a larger extent by 25.0% and 37.2%. Taken together, these results show that the policy, by making recreational cannabis massively accessible, encourages users to experience illicit drugs at a level that is more than the desire to try recreational cannabis.

What is the reasoning behind the policy's gateway effect? If the increased number of users who initiated and used illicit drugs is indeed due to mass access to recreational cannabis, we should see a surge in actual transactions of illicit drugs. The illicit drug trade is thriving on darknets because they are seen as much safer and more profitable than street dealing. The darknets offer vital anonymity for sellers and buyers who use cryptocurrencies such as Bitcoin to process illicit drug transactions. We investigate whether the policy led to more illicit drug transactions using Bitcoin on darknets. Under the gateway effect, we predict that darknet-related Bitcoin transactions will surge after the policy. For a complete picture of Bitcoin transactions, we focus on the change in both the transaction value and the transaction volume after the policy. Intriguingly, our *third* set of findings shows that the transaction value of darknet-related Bitcoins and the transaction volume surged by 163.0% and 64.7%, respectively, in the policy-affected states. The results suggest that, after the policy, users not only showed a higher desire to try illicit drugs but also made more purchases of illicit drugs from darknets, which is a critical reasoning behind the policy's gateway effect.

If the policy magnifies drug use by encouraging users to desire and purchase illicit drugs, we should also observe increased overdose mortality in policy-affected states, especially those allowing a more lenient use of recreational cannabis by individuals. Our *fourth* set of findings demonstrates that, in states legalizing recreational cannabis, for every additional ounce of recreational cannabis allowed, the number of illegal drug-related overdose mortality increased by 72%. Besides, for each 10% lower in the excise tax rate, the effect of the policy on illicit drug mortality is 15.51% higher. These findings suggest that the gateway effect became more salient in states with a more lenient policy on recreational cannabis. Such leniency in policy facilitated the spread of addiction to illicit drugs and worsened overdose mortality.

Our study fills several voids in the literature. First, most studies focus primarily on opioids (Hsu and Kovács, 2021; Mathur and Ruhm, 2023; Sabia et al., 2021), with less attention on illicit drugs that can pose larger harm to public health. Our study contributes to the literature by investigating RCL's impact on illicit drugs in a full spectrum, from the illicit drug trades and drug use to overdose mortality. Particularly, our novel data on darknet transactions adds less explored evidence on illicit drug trades. Second and more importantly, prior research does not discuss the underlying reasoning of the impact of RCL (Hsu and Kovács, 2021) or only speculate the reasoning without empirical evidence (Chan et al., 2020; Mathur and Ruhm, 2023; Sabia et al., 2021). Our study sheds light on the reasoning behind the policy impact for actionable policymaking. Taken together, our study addresses several data limitations in prior literature with much-needed evidence that informs the reasoning behind the RCL's impact.

Our research provides important implications for policymaking. We contribute timely evidence that addresses the long-debated effect of legalizing recreational cannabis. We reveal a significant gateway effect of the policy on illicit drug use and drill down to the multifaceted reasonings behind such an effect. When deciding whether to legalize recreational cannabis, policymakers should be aware of the gateway effect and the tradeoffs between enabling mass access to recreational cannabis and its consequences on public health. For policymakers in states passing the policy, formulating auxiliary measures to strengthen illicit drug controls becomes increasingly pressing. Future evaluations of legalization should monitor, as evident in our study, the darknet market trades of illicit drugs, the leniency of the policy, and the inequality of policy impacts among population groups. Specifically, by showing the importance of darknet markets as a critical source of illicit drugs in facilitating the policy's gateway effect, our study calls for immediate law enforcement resources and actions to monitor and crack down on illicit drug trades, especially those trades with traceable cross-border IP addresses. To this end, law enforcement should be equipped with technologies that can leverage deanonymization techniques to identify illicit drug-related transactions and combat cyber-trades of illicit drugs. Additionally, a cautious cap on the dosage allowance and a higher excise tax rate for recreational cannabis should be considered and, more importantly, strictly enforced in all licensed channels, such as local dispensaries and retail outlets. Only if we achieve a better understanding of these findings can policymakers and the public make informed decisions on regulations that benefit public health without risking the wellness of certain groups. To conclude, policymakers who propose to legalize and regulate recreational cannabis use need to fund research to monitor the impacts of these policy changes on public health and take advantage of this research to develop ways of regulating recreational cannabis use that minimize adverse effects on public health.

The remainder of the paper is organized as follows. In the next section, we summarize the related literature. Section 3 describes the data and measures. In Section 4, we present our empirical strategies and results. Section 5 concludes the paper. Finally, we exhibit all additional tests and discussions that support the main results in the Web Appendix.

2 Related Literature

2.1 Medical Cannabis and Healthcare

Cannabis is the most common illicit drug in the world and one of the most abused substances in the United States, following alcohol and tobacco (Zehra et al., 2018). However, when used appropriately, cannabis has medical benefits such as chronic pain relief and life-threatening condition treatment (Ellis et al., 2009; Lau et al., 2015). As a result, certain states in the United States launched policies to allow cannabis use for medical purposes. Medical cannabis legalization allows the production, sale, and use of cannabis when it is limited to medical purposes and no supply to commercial markets is involved (Caulkins et al., 2016). In 1996, California became the first state in the United States to legalize medical cannabis, followed by Washington, Oregon, Alaska, and Nevada. To date, medical cannabis legalization has spread to 34 states in the United States (Mahabir et al., 2020). As of 2020, the market of medical cannabis has garnered around \$7.3 billion in revenue with a 10.5% annual growth rate,[3] and cannabis has been widely used to treat various medical conditions such as cancer, neurology, HIV/AIDS, and muscle spasms (Urits et al., 2019).

Despite the benefits, medical cannabis is not free of damaging effects on the patients' bodies. CDC warns that medical cannabis can damage one's lungs, cardiovascular system, and brain.[4] Clinical evidence also shows that over-ingestion of cannabis can cause acute psychological distress, severe hyperemesis, and even mortality in extreme cases (Matheson and Le Foll, 2020). Thus, the disadvantages could outweigh the advantages of using cannabis as a medicine. Such risks raise public concerns that legalizing cannabis for medical purposes may increase harm to patients and become a hazard to public health.

In addition to its potential harm to patients, researchers have also raised concerns about medical cannabis on the phenomenon of medical cannabis diversion, whereby prescription cannabis is diverted to and abused by someone without a prescription (Thurstone et al., 2011). This phenomenon has become very common and has resulted in a strong dependence on cannabis in both patients and non-patients, leading them to use cannabis more frequently and develop serious addiction (Nussbaum et al., 2015). To fulfill their needs, these users may rely on not only dispensaries but also illicit markets to obtain cannabis (Reed et al., 2021). Apart from cannabis, illicit drugs such as cocaine and heroin can also be obtained from the illicit markets, and research has identified the association between medical cannabis legalization and increased use of cocaine and heroin (Wong and Lin, 2019). Research has also shown that restrictions on opioid prescriptions may even lead to more heroin-related overdose deaths (Diwas et al., 2022). These phenomena deviate from the good faith of policymakers to legalize cannabis (even just for medical purposes) and can potentially cause significant harm to public health.

2.2 Recreational Cannabis Legalization in the United States

The prohibition of cannabis for recreational purposes has long been the dominant scheme in policymaking (Cox, 2018). Yet, such a restriction has been relaxed recently. Although recreational cannabis is still illicit at the federal level, an increasing number of states, such as Colorado and Washington, have legalized cannabis for recreational use (Wilkinson et al., 2016). The legalization of recreational cannabis provides mass access to cannabis. Individuals can easily purchase allowed doses at licensed dispensaries without worrying about legal consequences. With recreational cannabis gradually liberalized in the United States, the policy has boosted the fast growth of the cannabis industry and created immediate and significant economic impacts. In 2020, recreational cannabis sales in the United States hit a record \$17.5 billion as Americans consumed more cannabis than ever before, and it is predicted to reach \$41 billion by 2026, roughly the size of the entire craft beer industry. Relatedly, excise and sales taxes on recreational cannabis in policy-affected states surpassed \$3 billion in 2020.[5] These numbers speak to the phenomenal economic impact of RCL.

Despite the evident economic benefits, the rapidly increasing availability of cannabis for recreational purposes aggravates public concerns. Yet, the policy implications for public health remain controversial and inconclusive. Supporters of RCL expect recreational cannabis, once legalized, to be better regulated to protect consumers and reduce adolescent access (Hall and Lynskey, 2016). Governments can also increase revenues by taxing the recreational cannabis industry and releasing resources from the justice system to address drug abuse and other crimes (Dragone et al., 2019; Felson et al., 2019; Leyton, 2019). More importantly, they argue the policy can help drug users to substitute illicit drugs (e.g., cocaine and heroin) with cannabis, which will mitigate the overdose epidemic and is beneficial to public health (i.e., the policy's substitution effect). Such claims are not groundless. Studies reveal that recreational cannabis legalization policies are associated with reductions in cigarette and alcohol use and can lead to fewer arrests and treatment related to heroin and cocaine (Alley et al., 2020; Calvert and Erickson, 2021; Choi et al., 2019; Chu, 2015; Hsu and Kovács, 2021; Liu and Bharadwaj, 2020). Besides, prior studies indicate that RCL is related to reduced opioid mortality (Ali et al., 2021; Chan et al., 2020; Livingston et al., 2017; McMichael et al., 2020; Sabia et al., 2021).[6] However, none of these studies have sought to pinpoint convincing evidence supporting the reduction of opioid-related mortality after the policy and the underlying reasonings behind such an effect. Additionally, the discussion of the policy impact on opioids does not speak to the same effect on illicit drugs, which cover overlapping but different drug categories. Literature has shown that recreational cannabis legalization does not stop people from engaging in illicit markets to obtain illicit drugs (e.g., Reed et al., 2021), and the reduction of opioid-related mortality does not necessarily mean the policy can also reduce the overdose mortality caused by illicit drugs.

In contrast to the supporter's view of RCL's substitution effect, critics insist that the legalization of recreational cannabis may, in fact, be associated with increased use of prescription drugs and even higher opioid mortality since RCL may become a gateway to illicit drugs (Alley et al., 2020; Caputi and Humphreys, 2018; Cruz et al., 2018; Lee et al., 2020; Mathur and Ruhm, 2023; Resko et al., 2019). They are concerned that the policy will increase the use of recreational cannabis, which will induce drug dependence and become a gateway to illicit drugs (i.e., encourage users to try illicit drugs), therefore worsening the overdose epidemic and endangering public health (Cleveland and Wiebe, 2008; Hall, 2015; Kandel et al., 1992; Olfson et al., 2018). The reasons behind this concern are mainly twofold. First, the psychological or physical gratification from using recreational cannabis diminishes over time, thereby prompting users to try illicit drugs to regain the initial level of gratification (DeSimone, 1998). Second, recreational cannabis users are often found to have higher curiosity and less concern about using illicit drugs (DeSimone, 1998; Kleiman, 1992). However, most extant studies focus solely on opioids to evaluate the impact of RCL but spend less attention on illicit drugs. Illicit drugs are defined differently from opioids and can pose higher harm to public health, especially when the gateway effect assumption holds. Although several studies take heroin and cocaine into consideration, other illicit drugs such as lysergide (LSD) and hallucinogens are overlooked (Hsu and Kovács, 2021; Mathur and Ruhm, 2023; Sabia et al., 2021). In addition, previous research analyzes illicit drug-related use and mortality but neglects illicit drug trades, a premise and critical stage of illicit drug use.

The debate on RCL's impacts, i.e., substitution and gateway, is of utmost importance to public health. As RCL is being passed across the United States, policymakers and the public should be aware of its unintended consequences, consider the tradeoffs, and implement the policy with precaution and necessary alteration. It is evident that, despite the considerable body of research, the literature does not conclude on the implications of the policy on public health. RCL's impacts on illicit drugs and, more importantly, the underlying reasonings of such impacts remain largely inconclusive. The voids in the literature motivate us to examine empirically how the policy affects the use of illicit drugs and the related overdose mortality, thereby adding much-needed evidence to the debate. Besides reconciling the debate with empirical evidence, we reveal novel insights on the reasonings (i.e., darknet transactions for illicit drugs, the leniency of recreational cannabis allowance and excise tax rates, and intention to use recreational cannabis) behind the policy impact for actionable policymaking, which are largely missing in prior literature. As a result, our research advances the understanding of why and how RCL impacts illicit drug use.

2.3 Darknet, Bitcoin, and Illicit Drugs

Illicit drugs, including cannabis before its legalization, are traditionally supplied by drug trafficking organizations that employ couriers to smuggle drugs from source countries into host countries (Kenney, 2007). In this process, illicit drugs are sold through underground distribution networks and by street dealers (Desroches, 2007), where vendors and buyers face multiple risks (coercion, violence, arrest, exposure to other drugs, etc.).

The rise of darknet markets, an encrypted layer of the internet where criminals conduct their business freely, has greatly changed the illicit drug trade practice. Due to expanding internet availability, evolving technologies and the profusion of social media have profoundly changed the delivery of healthcare (Liu et al., 2024; Zhou et al., 2022). The sales of illicit substances also enjoy dramatic growth on darknet markets (Buxton and Bingham 2015). Particularly, the United States leads other countries as the number one origin and destination country for illicit drug deals on darknet markets (Dolliver, 2015). Compared with street sales, darknet markets leverage anonymous shopper browsing and contactless drug delivery (usually by mail) to eliminate physical interactions between transacting parties and, consequently, the risk of illicit drug trades (Broséus et al., 2016). With sophisticated, user-friendly, and increasingly secure technologies, such hidden markets present a much safer environment than offline streets for illicit drug transactions. These benefits obviously infused the popularity of trading illicit drugs, which account for a primary product category on darknet markets (Broséus et al., 2017). According to the World Drug Report 2023 of the United Nations Office on Drugs and Crime (UNODC)[7], illicit drugs accounted for over 90% of sales on darknet markets.

Bitcoins and other similar cryptocurrencies have played an important role in facilitating darknet drug trades. While Bitcoin transactions are public, Bitcoin ownership is anonymous. Each Bitcoin is stored in a digital wallet identifiable only by a unique string of numbers and letters. Users do not always have to provide personal information to sign up for a Bitcoin wallet. Since Bitcoin provides confidentiality, it has been extensively used as the primary currency of choice for illicit drug traffickers in darknet transactions since 2011, when a famous darknet market, the Silk Road, emerged (Kethineni et al., 2018; Trautman, 2014). It is estimated that 46% of Bitcoin transactions on darknet markets are involved in illegal activities, such as illicit drug trades, valued at around $76 billion per year (Foley et al., 2019).

In line with this, recent studies have specifically pointed out that high-risk illicit drugs like heroin and cocaine are predominantly bought and sold on darknets through Bitcoin transactions (Broadhurst et al., 2020; Gilbert and Dasgupta, 2017; Lamy et al., 2021; Maras et al., 2023). The consensus among these studies strengthens our understanding that Bitcoin transactions are closely linked to illicit activities, particularly involving illicit drugs. This new form of illicit drug trading method poses a major challenge to law enforcement agencies. For years, authorities have been playing cat-and-mouse games with darknet users, but darknet drug trades continue to thrive (Van Buskirk et al., 2017).

Although Bitcoin transactions are anonymized in the form of user identity, its confidentiality cannot be guaranteed and can be hacked. Since all transactions are permanently and publicly stored in the Bitcoin network, similar transactions can be clustered to reveal a user's activities. Given the peer-to-peer nature of the Bitcoin network, a user's IP address can be logged by connected monitoring nodes, and the real-world identities of the user can be inferred through technical approaches (Fanti and Viswanath, 2017; Juhász et al., 2018; Koshy et al., 2014). An emerging research stream has focused on developing techniques to deanonymize Bitcoin transactions by identifying Bitcoin users' geolocations (Bartoletti et al., 2018; Liang et al., 2019; Monamo et al., 2016), thereby monitoring Bitcoin users and curbing related illegal activities. Frontier industry practitioners such as BlockCypher can monitor the flow of Bitcoins between different IP addresses and match the IP addresses with known wallets that are collected from darknets, social media, and various sources. By setting up a large number of monitoring nodes in the Bitcoin network, the source of a Bitcoin transaction and the counterparties' geolocations can be further identified by analyzing the transaction's relay trace. The service provides researchers with the opportunity to investigate illicit drug trades on darknet markets and becomes the source of data in our study.

3 Data

Legalizing mass access to recreational cannabis by U.S. states provides a unique empirical opportunity to examine the implications of recreational cannabis to public health, specific to the overdose mortality of illicit drugs. Table 1 shows the states that legalized recreational cannabis as of the end of 2018. As an overview of our data sources, we first collected the number of overdose mortality of illicit drugs, along with legal drugs (i.e., prescription opioids, as mentioned above) and alcohol, from the Centers for Disease Control and Prevention (CDC). To explore the reasonings behind the policy impact, we also collect three sources of data: the number of people who initiated and used illicit drugs, along with recreational cannabis and legal drugs, from the National Survey on Drug Use and Health (NSDUH)[8], the data of Bitcoin transactions on darknet markets from three service providers that analyze the Bitcoin network (i.e., WalletExplorer.com, BlockCypher.com, and IPInfo.io), and the data on the dosage allowance and excise tax rates of recreational cannabis from RCL bills in the policy-affected states. To support the main findings, we further collect a plethora of additional data sources such as the Treatment Episode Data Set: Admissions (TEDS-A), a national data system of annual admissions to substance use treatment facilities, the state-level Drug Enforcement Administration (DEA) drug seizures, the state-level Prescription Drug Monitoring Program (PDMP) and the Medical Cannabis Legalization (MCL) from the Prescription Drug Abuse Policy System (PDAPS). We complement the above data with a rich set of economic and demographic characteristics from the U.S. Census Bureau's annual American Community Survey (ACS) and the general search interest of Bitcoin, darknet, and illicit drugs on Google.

Table 1. Rollout of RCL across the U.S. states (2012–2018).

Effective date	State
2018/12/6	Michigan
2018/7/1	Vermont
2017/1/30	Maine
2017/1/1	Nevada
2016/12/15	Massachusetts
2016/11/9	California
2015/7/1	Oregon
2015/2/26	District of Columbia
2015/2/24	Alaska
2012/12/10	Colorado
2012/12/6	Washington

We merge all the data sources at the state-by-year-month level, the unit of analysis in our analyses.

For full availability of the data, our study period is between January 2012 and December 2018,[9] during which 11 states have legalized recreational cannabis. There were altogether 120,414 overdose mortality cases caused by illicit drugs in these policy-affected states. Table 2 presents the definition and summary statistics of the main variables in this study. We take logarithms of several variables with skewed distributions and denote them using log(*) throughout the paper. In the following subsections, we explain each of our data sources in more detail.

3.1 Overdose Mortality of Illicit Drugs

We collected data on the overdose mortality of illicit drugs, measured by log(illicit drug mortality) from the Multiple Cause of Death data from the CDC's Wide-ranging Online Data for Epidemiologic Research (WONDER) database, which is a major source of public health information (e.g., mortality, births, and vaccinations) in the United States (Friede et al., 1993). Despite our interest in investigating the impact of RCL on the overdose mortality of illicit drugs, we also collected data on the overdose mortality of legal drugs, measured by log(legal drug mortality$_{it}$), and the overdose mortality of alcohol, measured by log(alcohol mortality$_{it}$), in order to check the general trends of mortality from substance abuse. Specifically, we first requested from WONDER the overdose mortality data for U.S. residents of all demographics (by age, gender, origin, and race) at the state-by-year-month level. WONDER reports the number of deaths by cause in accordance with the T40 Codes,[10] which specify the drug categories that induced the overdose mortality (e.g., heroin-induced overdose mortality). Our approach of querying the data on the causes of death in WONDER follows the official guide of the CDC and has been widely used in previous research (Powell et al., 2018; Team CPDO, 2013).

Table 3 exhibits drugs in the T40 codes. By the cause of overdose mortality, we divided these drugs into four groups: illicit drugs, legal drugs, cannabis, and others. Specifically,

Table 2. Variable definitions and summary statistics.

Variable	Definition	Mean	SD	Min	Max	Period
The Policy						
Treated_i	Whether state *i* has enacted RCL in our sample	0.22	0.41	0.00	1.00	2012–2018
$\text{Treated}_i \times \text{Post}_{it}$	Whether state *i* has enacted RCL by month *t*	0.09	0.29	0.00	1.00	
CDC Wide-ranging Online Data for Epidemiologic Research (WONDER)						
$\log(\text{illicit drug mortality}_{it})$[19]	Log of the number of overdose mortality caused by illicit drugs in state *i* and month *t*	1.94	1.80	0.00	5.44	2012–2018
$\log(\text{cannabis mortality}_{it})$	Log of the number of overdose mortality caused by cannabis in state *i* and month *t*	0.04	0.20	0.00	2.20	
$\log(\text{legal drug mortality}_{it})$	Log of the number of overdose mortality caused by legal drugs in state *i* and month *t*	2.72	1.73	0.00	6.02	
$\log(\text{alcohol mortality}_{it})$	Log of the number of overdose mortality caused by alcohol in state *i* and month *t*	3.31	1.38	0.00	6.25	
$\log(\text{illicit drugs} + \text{T40.4} + \text{T40.6 mortality}_{it})$	Log of the number of overdose mortality caused by illicit drugs and T40.4 + T40.6 in state *i* and month *t*	2.73	1.83	0.00	6.11	
$\log(\text{T40.4} + \text{T40.6 mortality}_{it})$	Log of the number of overdose mortality caused by T40.4 + T40.6 in state *i* and month *t*	1.84	1.84	0.00	5.93	
$\log(\text{T40.1} + \text{T40.5 mortality}_{it})$	Log of the number of overdose mortality caused by T40.1 + T40.5 in state *i* and month *t*	2.20	1.82	0.00	5.48	
$\log(\text{unspecified mortality}_{it})$	Log of the number of unspecified causes of mortality in state *i* and month *t*	3.03	1.78	0.00	6.73	
National Survey on Drug Use and Health (NSDUH)						
$\log(\text{illicit drug initiation}_{it})$	Log of the number of people who initiated illicit drugs in state *i* and every two years *t*	9.14	1.16	6.91	12.25	2013, 2015–2018
$\log(\text{cannabis initiation}_{it})$	Log of the number of people who initiated cannabis in state *i* and every two years *t*	10.32	1.03	8.01	13.00	
$\log(\text{legal drug initiation}_{it})$	Log of the number of people who initiated pain relievers in state *i* and every two years *t*	4.36	0.49	3.00	5.41	2016–2018
$\log(\text{illicit drug use}_{it})$	Log of the number of people who used illicit drugs in state *i* and every two years *t*	4.72	0.45	3.70	5.78	2016–2018
$\log(\text{cannabis use}_{it})$	Log of the number of people who used cannabis in state *i* and every two years *t*	5.49	0.51	4.48	7.43	2013, 2015–2018
$\log(\text{legal drug use}_{it})$	Log of the number of people who misused pain relievers in state *i* and every two years *t*	4.60	0.46	3.48	5.61	2016–2018

(continue)

Table 2. *Continued.*

Variable	Definition	Mean	SD	Min	Max	Period
Treatment Episode Data Set: Admissions (TEDS-A)						
log(admission_noprior$_{it}$)	Log of the number of records without prior substance use in state *i* and every year *t*	8.28	2.47	0.00	11.89	2012–2018
log(admission_prior$_{it}$)	Log of the number of records with prior substance use in state *i* and every year *t*	8.09	1.73	0.00	10.77	2012–2018
log(admission_heroin$_{it}$)	Log of the number of records with heroin reported at admission in state *i* and every year *t*	7.92	1.72	3.53	11.59	
log(admission_cannabis$_{it}$)	Log of the number of records with cannabis reported at admission in state *i* and every year *t*	8.82	1.11	6.02	11.67	
log(admission_cocaine$_{it}$)	Log of the number of records with cocaine reported at admission in state *i* and every year *t*	7.80	1.48	3.71	11.57	
Drug Enforcement Administration (DEA)						
log(seizure$_{it}$)	Log of the number of drug seizures in state *i* and every year *t*	9.37	1.33	4.55	11.70	2012–2018
Bitcoin Transactions						
logBTVOL$_{it}$	Log of the number of Bitcoin transactions in state *i* and month *t*	5.22	2.10	0.69	10.06	Dec 2014-Dec 2015
logBTVAL$_{it}$	Log of Bitcoin transaction value in USD in state *i* and month *t*	12.49	2.65	1.61	17.63	
Google Trends						
GTBitcoin$_{it}$	Search interest in the keyword “Bitcoin” on Google in state *i* and month *t*	48.41	18.56	11.00	100.00	Dec 2014-Dec 2015
GTDarknet$_{it}$	Search interest in the keyword “Darknet” on Google in state *i* and month *t*	34.53	20.39	0.00	100.00	
GTHeroin$_{it}$	Search interest in the keyword “Heroin” on Google in state *i* and month *t*	51.55	18.97	21.00	100.00	
GTCocaine$_{it}$	Search interest in the keyword “Cocaine” on Google in state *i* and month *t*	76.11	13.75	34.00	100.00	
State Bills on RCL						
AddDosage$_{i}$	The dosage allowance of recreational cannabis in state *i* subtracted with 1	−0.90	0.36	−1.00	1.50	2012–2018
ExciseTax$_{i}$	The excise tax rate of recreational cannabis in state *i*	0.10	0.11	0.00	0.37	
State Drug Monitoring Programs						
PDMPPost$_{it}$	Whether state *i* enacted the Prescription Drug Monitoring Program (PDMP) in our sample	0.94	0.24	0.00	1.00	2012–2018
MCLPost$_{it}$	Whether state *i* enacted the medical cannabis legalization (MCL) in our sample	0.46	0.50	0.00	1.00	
American Community Survey (ACS)						
log(MI$_{it}$)	Log of the median household income in state *i* and year *t*	10.92	0.17	10.57	11.32	2012–2018
Gini Index$_{it}$	A coefficient that demonstrates a degree of income inequality in state *i* and year *t*	0.47	0.02	0.41	0.56	
%BD$_{it}$	Percentage of the population with a bachelor’s degree in state *i* and year *t*	0.13	0.02	0.08	0.19	

Table 3. Drugs that cause overdose mortality in the WONDER ICD-10's T40 category.

Cause of death	Code	Drug name	Definition
Illicit drugs	T40.1	Heroin	One of the world's most dangerous opioids
	T40.5	Cocaine	A powerfully addictive stimulant drug made from the leaves of the coca plant
	T40.8	Lysergide [LSD]	One of the most powerful mind-altering chemicals
	T40.9	Other and unspecified psychodysleptics [hallucinogens]	Drugs that alter a person's awareness of their surroundings, thoughts, and feelings, including two categories: classic hallucinogens and dissociative drugs
Legal drugs	T40.2	Other opioids	Natural opioid analgesics and semi-synthetic opioids
	T40.3	Methadone	A synthetic opioid
	T40.4	Other synthetic narcotics	Synthetic opioid analgesics other than methadone, including drugs such as fentanyl and tramadol
Cannabis	T40.7	Cannabis (and derivatives)	A psychoactive drug from the Cannabis plant used primarily for medical or recreational purposes
Others[20]	T40.0	Opium	Dried latex obtained from the seed capsules of the opium poppy
	T40.6	Other and unspecified narcotics	Drugs in T40 that exclude the previous categories

illicit drugs (T40.1, T40.5, T40.8, and T40.9) are those drugs that have higher risks of causing physical and psychological addiction and death (Atkins et al., 2019). Legal drugs (T40.2, T40.3, and T40.4) are prescription opioids and can be legally obtained with medical prescriptions, but they are still controlled substances that may lead to drug addiction (Powell et al., 2018). Since cannabis (T40.7) is legal in policy-affected states but remains illicit in other states without the policy, we list cannabis (i.e., T40.7) separately as an independent drug type.

3.2 Initiation and the Use of Illicit Drugs

As part of the investigation of the reasonings behind the RCL effect, we collected data on illicit drug initiation and use from the National Survey on Drug Use and Health (NSDUH), sponsored by the Substance Abuse and Mental Health Services Administration (SAMHSA). Our goal is to examine whether there are indeed more users initiating and using illicit drugs after the policy, an important channel for the policy effect to manifest. The NSDUH contains annual survey data about the use of illicit drugs, alcohol, and tobacco across the states in the United States and is widely used for research on illicit drugs (Azofeifa et al., 2019).[11] We extracted "Illicit Drug Other Than Marijuana Dependence or Abuse—Past Year (UDPYIEM)" from NSDUH to observe the number of people who used illicit drugs other than cannabis, i.e., log(illicit drug use$_{it}$). We also extracted the "Past Year Initiation of Cocaine Use (RECCOC2)" to proxy the number of people who initiate illicit drug use, denoted by log(illicit drug initiation$_{it}$).[12]

In addition to the initiation and use of illicit drugs, we also collected the initiation and use for cannabis, log(cannabis use$_{it}$), and log(cannabis initiation$_{it}$), from NSDUH's "Marijuana—Past Month Use (MRJMON)" and "Past Year Initiation of Marijuana Use (RECMJ2)." The purpose is to validate the premises of RCL, i.e., whether the policy leads to increased cannabis initiation and use as suggested in prior research (Cerdá et al., 2020; Kerr et al., 2017; Miller et al., 2017; Rusby et al., 2018). We further collected the initiation and use data of legal drugs, log(legal drug initiation$_{it}$), and log(legal drug use$_{it}$) from NSDUH's "Past Year Initiation of Pain Reliever Misuse (RECPNRNM)" and "Pain Relievers—Past Month Misuse (PNRNMMON)."[13] Similar to the overdose mortality of alcohol, the purpose of collecting legal drug initiation and use data is to determine whether a general trend of addictive substance use would drive the policy impacts in causing the increased overdose mortality of illicit drugs.

3.3 Bitcoin Transactions on Darknet Markets

To investigate the underlying reasonings behind the RCL effect further, we focus on Bitcoin transactions on darknet markets, a major source of illicit drugs. We begin with collecting Bitcoin transaction data from 17 major darknet markets in the United States, including the famous Silk Road and AlphaBay, along with other active darknet sites such as Evolution and Nucleus. Our data source is WalletExplorer.com, a blockchain explorer that tracks Bitcoin wallets that primarily belong to darknet markets, gambling websites, and cryptocurrency exchanges, which is widely used in the literature (Foley et al., 2019; Jourdan et al., 2018). We obtained a total of 5,188,703 transactions of 2,325,757 Bitcoin wallets that are active receivers (i.e., the buyers' paying accounts) on the 17 darknet markets. Table A3 in the Web Appendix exhibits the full list of these 17 darknet markets and their lifespan (from launch to close), as well as the numbers of Bitcoin transactions and wallets for each market.

Although the ownership of Bitcoin is anonymous, these transactions are publicly visible. We further traced the IP addresses of each Bitcoin transaction from the wallets using BlockCypher's Transaction API services. BlockCypher is one of the largest Bitcoin and blockchain web service platforms that provide various analytical insights on Bitcoin transactions, such as relay IP addresses, timestamps, and transaction value (Brinckman et al., 2017). Bitcoin leverages a gossip-style protocol in which nodes in the blockchain relay messages across the network.[14] Thanks to BlockCypher's large number of monitoring nodes deployed in the Bitcoin network, we were able to obtain from its Transaction API service the IP address of the first Bitcoin node that relays a specific transaction in the Bitcoin network, which corresponds to a "relayed_by" attribute in each Bitcoin transaction. From the relay IP recorded by BlockCypher, we were able to identify the unique user initiating the Bitcoin transaction (i.e., the illicit drug buyer in our context). After obtaining the IP address of each Bitcoin transaction, we inferred the geolocations of the illicit drug buyer using the IP address information lookup API service of IPInfo.io, a leading IP address data provider. Combining the Bitcoin transactions and the geolocation of associated IP addresses, we calculated the volume ($logBTVOL_{it}$) and total value ($logBTVAL_{it}$) of Bitcoin transactions by state and year-month. Note that because BlockCypher did not have the monitoring nodes deployed until December 2014, the IP addresses of Bitcoin transactions on 6 darknet markets (SilkRoad, BlueSky, Pandora, Sheep, SilkRoad2, and CannabisRoad) are not obtainable and thus omitted, leaving 11 darknet markets with both Blockchain wallets and IP addresses of each transaction in our sample.

We complemented the Bitcoin transaction data with the search interest of the keyword "Bitcoin" on Google Trends data. Over the past few years, Bitcoin has drawn increasing attention from the public and experienced rapid growth in transactions. We leveraged the measure of search interest in "Bitcoin," denoted by $GTBitcoin_{it}$, to test whether the impacts are driven by a general trend of Bitcoin rather than the policy. Specifically, the $GTBitcoin_{it}$ is on a scale from 0 to 100, with 100 indicating that in the state; "Bitcoin" was the most popular search term among all Google searches in that month.

3.4 Dosage Allowance and Excise Tax Rates Across States

Although recreational cannabis is legalized in the policy-affected states, there is variation across these states in terms of the maximum dosage of recreational cannabis that a user is allowed to possess, as well as in the excise tax rates applied to recreational cannabis. The variance in both dosage allowance and excise tax rates is indicative of the states' leniency in the operations of the RCL policy. To nail down the reasonings behind the RCL effect, we are interested in investigating whether overdose mortality of illicit drugs would vary by the dosage allowance and excise tax rates of recreational cannabis in the policy-affected states. Specifically, in states with a higher dosage allowance of recreational cannabis (i.e., more lenient), if we observe higher overdose mortality of illicit drugs, the evidence will support the gateway effect; in contrast, if we observe lower overdose mortality of illicit drugs after the policy, the evidence will speak to the substitution effect.

Similarly, drawing from prior research, excise taxes have been effective in reducing the consumption of addictive substances like tobacco and alcohol, thereby lessening their adverse health impacts (Chaloupka et al., 2019). Thus, if we observe higher overdose mortality from illicit drugs in states with lower excise tax rates on recreational cannabis (indicative of a more lenient policy approach), it would lend support to the gateway effect hypothesis. It suggests that more lenient RCL policies might inadvertently enhance access and affordability of cannabis, increasing the likelihood that individuals may transition to more harmful illicit drugs, thereby intensifying the gateway effect.

To examine the predictions, we collected the dosage allowance and excise tax rate data by hand coding the information from RCL bills passed by all the 11 affected states. For example, for the state of California, the RCL bill titled "California Proposition 64" states that "adults are allowed to legally possess and use up to 28.5 grams (around 1 ounce) of cannabis for recreational purposes" and "a marijuana excise tax shall be imposed upon purchasers of marijuana or marijuana products sold in this state at the rate of 15 percent." We extracted such dosage allowance and excise tax rate information (e.g., 1 ounce and 15%) for all the 11 policy-affected states, as exhibited in Table 4.

It appears that the majority of the states allow a user to possess up to 1 ounce of recreational cannabis at a time, whereas states such as Michigan, Maine, and the District of Columbia are more lenient in the dosage allowance (up to 2.5 ounces). Thus, we constructed a state-specific variable, $AddDosage_i$, wherein we subtracted one from the dosage allowance of recreational cannabis in every state. This variable allows us to use 1 ounce as a benchmark to test the impact of the additional dosage allowance of recreational cannabis. In contrast, the excise tax rates exhibit more variation across RCL states. To examine how these rates might moderate the RCL's impact, we employ another state-specific variable, $ExciseTax_i$, which captures the excise tax rate for each state.

4 Results

As a quick roadmap, we begin by discussing the main specification of our empirical model in Section 4.1. We estimate the policy impacts on the overdose mortality of illicit drugs and report the policy impact (substitution versus gateway) in Section 4.2. In Section 4.3, we explore multiple sets of evidence supporting the reasonings behind the policy impacts. We further rule out alternative explanations and conduct a series of robustness checks in Section 4.4.

4.1 Empirical Specification

The variation in the RCL rollout across states at different times, as shown in Table 1, offers a unique quasi-experimental opportunity to determine the policy impacts (substitution versus gateway). Our empirical strategy is the standard DID

Table 4. Dosage allowance and excise tax of recreational cannabis in RCL-affected states.

State	Dosage allowance (up to)	Excise tax rate (%)	RCL bill[21]
Michigan	2.5 ounces	10	Proposal 1 of 2018
Vermont	1 ounce	14	H.511 (Act 86)
Maine	2.5 ounces	0	Question 1
Nevada	1 ounce	10	Question 2
Massachusetts	1 ounce	10.75	Question 4
California	1 ounce	15	Proposition 64
Oregon	1 ounce	0	Measure 91
District of Columbia	2 ounces	0	Initiative 71
Alaska	1 ounce	0	Measure 2
Colorado	1 ounce	15	Amendment 64
Washington	1 ounce	37	Initiative 502

Table 5. Comparability between the affected and unaffected groups.

	11 Affected states		40 Unaffected states			
	Mean	SD	Mean	SD	Mean difference	p
CDC WONDER						
log(illicit drug mortality$_{it}$)	1.373	0.211	1.359	0.281	0.013	0.741
log(cannabis mortality$_{it}$)	2.842	7.685	3.686	8.499	0.844	0.768
log(legal drug mortality$_{it}$)	1.515	0.262	1.471	0.265	0.044	0.185
log(alcohol mortality$_{it}$)	2.894	0.805	2.866	0.640	0.027	0.906
ACS						
log(MI$_{it}$)	4.723	0.051	4.688	0.070	0.036	0.122
Gini Index$_{it}$	0.459	0.030	0.455	0.019	0.003	0.677
%BD$_{it}$	0.106	0.052	0.102	0.039	0.004	0.796

approach capitalizing on the staggered policy rollout. Our sample contains 51 states, including 11 states in the "affected group" that have enacted RCL and 40 states in the "unaffected group" that have not legalized recreational cannabis during our study period. Table 5 compares the affected and unaffected groups based on the mean values of all observed pre-treatment characteristics before the policy was first introduced in 2012 in the State of Washington.[15] Overall, we can see the covariates are quite balanced, and the two groups of states are not significantly different in terms of the observed pre-treatment characteristics. Although the mean cannabis mortality differs between the two groups, they are still not significantly different due to the large standard deviation.

Using the sample, the main specification of our empirical model is a standard difference-in-differences (DID) regression. Our main unit of analysis is at the state-by-year-month level[16]. The equation is as follows,

$$Y_{it} = \beta \cdot Treated_i \times Post_{it} + \mu_i + \nu_t + \varepsilon_{it}, \qquad (1)$$

where Y_{it} is the dependent variable. Depending on the specific regression, we examine the policy's impact on various outcomes (overdose mortality of illicit drugs, initiation and use of illicit drugs, etc.). We include state fixed effects, μ_i, and year-by-month fixed effects, ν_t, as in standard DID designs. The policy dummy, $Treated_i \times Post_{it}$, is the key regressor, and its estimated coefficient will capture the policy effect on the specific outcome (dependent variable). Note that the policy dummy is a state-by-year-month specific: it takes the value of 1 for a state affected by the policy and its given year-month is after the passing of RCL; it equals 0 elsewhere, including the unaffected states in all periods. ε_{it} is the error term.

4.2 The Policy Impact on Illicit Drug Mortality

As discussed previously, our main analysis explores RCL's impacts on multiple outcomes of illicit drug use to determine the primary effect of the policy, be it the substitution effect or the gateway effect. The gateway effect indicates that RCL, by providing mass access to recreational cannabis, provides a gateway for users to try illicit drugs. In other words, if we find that the number of overdose mortality from illicit drugs increases after the policy, the evidence may suggest the gateway effect. However, if the substitution effect dominates, the policy will likely introduce recreational cannabis as a substitute for illicit drugs, resulting in a decrease in overdose mortality.

We thus examine the policy's impact on the overdose mortality of illicit drugs to determine which effect dominates. Table 6 reports the regression results, in which we show results

Table 6. Impact of RCL on the overdose mortality of illicit drugs.

D.V.: log(illicit drug mortality$_{it}$)	(1)	(2)	(3)	(4)
$Treated_i \times Post_{it}$	1.110***	0.957***	0.679***	0.346***
	(0.112)	(0.072)	(0.119)	(0.055)
R^2	0.02	0.84	0.06	0.87
Observations	4284	4284	4284	4284
State FE	No	Yes	No	Yes
Year-by-month FE	No	No	Yes	Yes

Robust standard errors clustered at the state level are shown in parentheses. This analysis encompasses a 7-year period from 2012 to 2018, covering all 50 states and the District of Columbia. The unit of analysis is at the state-by-year-month level.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

of four specifications for robustness check purposes: an ordinary least square regression without fixed effects (column 1), two fixed-effect models with either state or year-month fixed effects (columns 2 and 3), and a fixed-effect model with state and year-month fixed effects (column 4).

The positive coefficients of the policy dummy in all regressions suggest that the policy increased the overdose mortality of illicit drugs, lending support to the gateway effect. Numerically, when the state and year-month fixed effects are controlled in column (4), the number of illicit drug mortality cases increased by about 34.6% (or 4.7 people[17]) per month in an average affected state. This effect size is startling, as it suggests that annually RCL results in 56.4 more illicit drug overdose mortality cases in an average affected state.

4.3 Evidence Behind the Policy Effects

Our main effect has shown the gateway effect of RCL in driving overdose mortality of illicit drugs. One natural question arises: what are the reasonings behind the policy's gateway effect? In this section, we make three predictions about the reasonings. First, if RCL indeed causes the increase in overdose mortality of illicit drugs, users should have shown an interest (tendency to initiate and use recreational cannabis and illicit drugs) after the policy. If the evidence supports that there are more users initiating and using recreational cannabis and illicit drugs after the policy, it would suggest the gateway effect.

Second, under the gateway effect, there should be increased activities of illicit drug transactions due to the increased demand for illicit drugs after the policy. Illicit drug transactions primarily occur on darknets (Du et al., 2019) and are facilitated by Bitcoin (Sun Yin et al., 2019). If there is a surge of Bitcoin transactions on darknets after the policy, the evidence would suggest its gateway effect.

Lastly, given that RCL is a localized policy at the state level, both the dosage allowance and the excise tax rate of recreational cannabis by the state should have played a role in influencing the extent of the gateway effect. That is, under the gateway effect, users in states with a lenient allowance can access a higher dose of recreational cannabis from licensed dispensaries, which may motivate them more to try illicit drugs and face a higher risk of overdose mortality (i.e., evidence for a stronger gateway effect). Similarly, states with lower excise tax rates might make recreational cannabis more affordable, possibly amplifying the gateway effect. In the following subsections, we examine each of these predictions.

4.3.1 Evidence from the Initiation and Use of Illicit Drugs.

We first test whether there are more users initiating and using recreational cannabis and, consequently, illicit drugs after the policy. Columns (1) and (2) in Table 7 report the regression results, with the dependent variable being log(cannabis initiation$_{it}$) and log(cannabis use$_{it}$), respectively. The specific measures are the estimated logged numbers of users who initiated recreational cannabis and users who used recreational cannabis in a state. We note that the coefficients of the policy dummies are positive and statistically significant. Such increases in initiations and use of cannabis speak about the premise of the policy, i.e., the RCL is effective in generating users' attempts to use recreational cannabis.

Although results in the first two columns do not necessarily suggest a substitution effect or a gateway effect of the policy, Columns (3) and (4) provide important evidence supporting the gateway effect. We find that, quantitively, the policy increased the number of users who initiate and use illicit drugs by 25.0% and 37.2%, respectively. This set of findings rejects the conjecture of a substitution effect and is consistent with the gateway effect. Taken together, these two sets of evidence in Table 7 reveal the reasonings behind the policy impact: by legalizing access to recreational cannabis, the policy creates access to more recreational cannabis and encourages users to seek the experience of using illicit drugs that are harder and stronger than cannabis (i.e., the gateway effect).

4.3.2 Evidence from Bitcoin Transactions on Darknet Markets.

We test the conjecture about the RCL effect by seeking another piece of evidence behind the RCL impact: the change in illicit drug transactions using Bitcoin on darknet markets. As darknet markets become a major source of trading illicit drugs, under the gateway effect, we should expect an increase in illicit drug transactions after the policy to meet the increased user demand for illicit drugs. Table 8 reports the regression that tests this prediction. We use the volume (or the number) of

Table 7. Impact of RCL on the initiation and use of cannabis and illicit drugs.

	(1)	(2)	(3)	(4)
D.V.s:	log(cannabis initiation$_{it}$)	log(cannabis use$_{it}$)	log(illicit drug initiation$_{it}$)	log(illicit drug use$_{it}$)
Treated$_i$ × Post$_{it}$	0.207***	0.091***	0.250*	0.372***
	(0.073)	(0.034)	(0.150)	(0.113)
R^2	0.97	0.99	0.91	0.98
Observations	255	255	255	153
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Robust standard errors clustered at the state level are shown in parentheses. The analysis for log(cannabis initiation$_{it}$), log(cannabis use$_{it}$), and log(illicit drug initiation$_{it}$) covers the years 2013, 2015, 2016, 2017, and 2018. The analysis for log(illicit drug use$_{it}$) covers the years 2016, 2017, and 2018. All analyses include data from all 50 states and the District of Columbia. The unit of analysis is at the state-by-year level.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

Table 8. Impact of RCL on bitcoin transactions of Darknet markets.

	(1)	(2)
D.V.s:	logBTVOL$_{it}$	logBTVAL$_{it}$
Treated$_i$ × Post$_{it}$	0.647***	1.630***
	(0.251)	(0.449)
R^2	0.91	0.83
Observations	635	635
State FE	Yes	Yes
Year-by-month FE	Yes	Yes

Robust standard errors clustered at the state level are shown in parentheses. This analysis covers the period from December 2014 to December 2015, encompassing data from all 50 states and the District of Columbia. The unit of analysis is at the state-by-year-month level.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

Table 9. Moderating effects of dosage allowance and excise tax of recreational cannabis.

D.V.: log(illicit drug mortality$_{it}$)	(1)	(2)
Treated$_i$ × Post$_{it}$	0.146**	0.465**
	(0.060)	(0.069)
Treated$_i$ × Post$_{it}$ × AddDosage$_i$	0.720***	
	(0.090)	
Treated$_i$ × Post$_{it}$ × ExciseTax$_i$		−1.551***
		(0.538)
R^2	0.87	0.87
N	4284	4284
State FE	Yes	Yes
Year-by-month FE	Yes	Yes

Robust standard errors clustered at the state level are shown in parentheses. This analysis encompasses a 7-year period from 2012 to 2018, covering all 50 states and the District of Columbia. The unit of analysis is at the state-by-year-month level.
*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

Bitcoin transactions (logBTVOLit) and the transaction value in USD (logBTVALit) in a state as two dependent variables. Interestingly, we find that both the volume and value of Bitcoin transactions on darknets increased significantly (more than 64.7% and 163.0% in the affected states relative to unaffected states) after the policy. Because none of the two policy effects on Bitcoin transactions were negative or remained unchanged, the evidence rejects the substation effect and lends further support to the gateway effect.

4.3.3 Evidence from the Dosage Allowance and Excise Tax Rates of Recreational Cannabis. If the gateway effect continues to hold, we expect that states with a more lenient allowance of recreational cannabis will experience a larger overdose crisis of illicit drugs as users possess more dosage of recreational cannabis (a stronger gateway to illicit drugs). To examine this effect, we modified Equation (1) into:

$$Y_{it} = \beta_1 \cdot Treated_i \times Post_{it} + \beta_2 \cdot Treated_i \times Post_{it} \times AddDosage_i + \mu_t + v_i + \varepsilon_{it}, \tag{2}$$

where β_2 captures the moderating effect of $AddDosage_i$ on the overdose mortality of illicit drugs. Note $AddDosage_i$ will be absorbed by the fixed effects v_i as it is state-specific and does not vary over time. As shown in column (1) of Table 9, we find that overdose mortality in illicit drugs increases in RCL-affected states, particularly those with a higher dosage allowance of recreational cannabis. Quantitively, for each additional ounce of dosage allowance, the impact of the policy on illicit drug mortality is 72% higher.

This evidence further supports the gateway effect, which is significantly more pronounced in states with a more lenient regime of RCL. Specifically, such leniency, by facilitating greater access to cannabis, has stimulated the interest of users in harder illicit drugs, causing increased overdose mortality and worsening the drug abuse epidemic. An important implication is that leniency in legalizing recreational cannabis may come at the cost of escalated overdose mortality from illicit drugs. The more lenient the policy, the greater the potential cost in terms of increased drug-related harm.

Similarly, if the gateway effect holds, we would also expect that states with lower excise tax rates on recreational cannabis, reflecting a more lenient RCL policy, would exhibit a higher overdose mortality of illicit drugs. To investigate this relationship, we adapted Equation (1) into:

$$Y_{it} = \beta_1 \cdot Treated_i \times Post_{it} + \beta_2 \cdot Treated_i \times Post_{it} \times ExciseTax_i + \mu_t + v_i + \varepsilon_{it}, \quad (3)$$

where β_2 captures the moderating effect of $ExciseTax_i$ on overdose mortality. As presented in column (2) of Table 9, we find that overdose mortality in illicit drugs is higher in policy-affected states with a lower excise tax rate on recreational cannabis. Specifically, for each 10% lower in the excise tax rate, the effect of the policy on illicit drug mortality is 15.51% higher. This suggests that, in states with lower excise taxes, where recreational cannabis is more affordable, the gateway effect is more pronounced. These findings lend support to the gateway effect by illustrating how variations in policy implementation, particularly in terms of taxation, can modulate the potential pathways to illicit drug use.

4.4 Additional Analyses

We have shown evidence supporting the gateway effect of RCL and the reasonings behind the effect. Several alternative explanations of the policy effect merit careful study. We first rule out these alternative explanations in Section 4.4.1. We then conduct robustness checks to triangulate the results and report the results in Section 4.4.2. As an overview, we find little evidence that these alternative explanations are drivers of the policy effect. The results from robustness checks are also qualitatively consistent with our main findings.

4.4.1 Excluding Alternative Explanations.

Trends of Substance Use. One may be concerned that the increases in illicit drug use and overdose mortality in our main results were not driven by the gateway effect, but there was a general trend of addictive substance abuse (e.g., legal drugs and alcohol) after the policy. If this is true, it means the policy impact is not directional (i.e., from cannabis to illicit drugs), and our assumptions of the RCL's gateway effect would be violated. This is a reasonable concern because the opioid epidemic in the United States has been a critical issue for many years (Manchikanti et al., 2012). Therefore, it is possible that illicit drugs became more popular after the policy because of a general trend rather than the gateway effect.

To examine this concern, we employed Equation (1) to test whether the policy also led to an increased initiation and use, as well as overdose mortality of addictive substances such as legal drugs and alcohol. Table A4 in the Web Appendix presents the policy impacts on the initiation and use of legal drugs. With all coefficients insignificant, we find no evidence that the policy increased the initiation and use of legal drugs. Table A5 in the Web Appendix reports the policy impacts on overdose mortality caused by legal drugs and alcohol. Again, we find no evidence that the policy led to death caused by legal drugs or alcohol. Together, the results in Tables A4 and A5 address the concern about the general trend of substance abuse in policy-affected states. These analyses further support our findings by showing that the policy did not promote the use of addictive substances (legal drugs, alcohol, etc.). Instead, the policy particularly increased the initiation, use, and overdose mortality of illicit drugs, as shown in Tables 6 and 7.

Trend of Bitcoin Popularity. We analyze the Bitcoin transactions on darknet markets to seek evidence supporting the reasonings behind the policy impacts (as in Section 4.3.2). One may argue that the increased volume and value of Bitcoin transactions are driven by a general trend of the increased popularity of Bitcoin rather than the policy. To examine this alternative explanation, we include the search interest of the keyword "Bitcoin" on Google for each state in a month as a control variable in the regression. As shown in Table A6 of the Web Appendix, we find the results are highly consistent with those in Table 8, continuing to support the reasonings of illicit drug transactions behind the gateway effect. In particular, the coefficients of $\text{GTBitcoin}_{\text{it}}$ suggest that the increased volume and value of Bitcoin transactions are driven by the policy instead of a general trend of Bitcoin popularity.

Trend of Fentanyl Use. Overdose deaths from fentanyl have become a growing problem in recent years. Since the timing of fentanyl's rise partially overlaps with our observation window, some may be concerned that our results may be driven by fentanyl, which is legal for medical use but frequently associated with illicit drug use and overdoses. To this end, we control the trend of potential fentanyl use in estimating the RCL impact on illicit drug mortality, using the overdose mortality caused by T40.4 and T40.6 as the control variable. As shown in Table A7 of the Web Appendix, the results do not take away our estimated impact of RCL and are consistent with our main findings.

State-level Law Enforcement Activities. One alternative explanation is that the increased use of illicit drugs could be due to reduced law enforcement activities in states with RCL. Exploring this alternative explanation would be important to establish the current gateway effect. To this end, we add the DEA (Drug Enforcement Administration) drug seizures—a proxy for state-level law enforcement—as controls into our model. If our previous findings are not driven by reduced law enforcement activities, adding these controls should not alter our previous findings. As shown in Table A8 of the Web Appendix, we find that the results are largely consistent with our main analyses, suggesting that the impact of RCL on illicit drugs is not driven by the reduced law enforcement activities.

State-level Drug Monitoring. There might also be concerns that co-occurring state-level policies such as the prescription drug monitoring program (PDMP) and medical cannabis legalization (MCL) may confound our results. Thus, we collect the implementation date data of PDMP and MCL from the Prescription Drug Abuse Policy System (PDAPS). We take into consideration PDMP and MCL as control variables, and the results, as shown in Table A9 of the Web Appendix, are consistent with our previous findings. In addition, we further add the state-time trend (i.e., for each state, we construct state-time trends by multiplying the month (from 1 to N) with the corresponding dummy variable for each state) to account for the differential state trend, and the results presented in Table A10 of the Web Appendix turn out to be consistent with our previous findings.

Cannabis Sales on Darknets. Some may be concerned that the increase in darknet drug sales after RCL is primarily driven by cannabis rather than illegal drugs, which can undermine the gateway effect. To address this concern, we label the darknet listings (Figure A1 illustrates an example) by drug types and demonstrate statistically that cannabis is not a major drug listing sold on darknets. Tables A11 to A13 present the statistical evidence. This further supports our main findings and the gateway effect.

4.4.2 Robustness Checks.

Relative Time Model. An important assumption of a DID design is that the affected group and unaffected group have relatively parallel trends before the treatment (i.e., the policy in our case). We leverage a relative time model (Angrist and Pischke, 2008) to check whether parallel trends exist between the affected and unaffected states prior to the introduction of RCL. The model is specified as:

$$Y_{it} = \sum_{\tau=-T}^{N} \lambda_\tau \cdot D_{it,\,\tau} + \mu_t + \nu_i + \varepsilon_{it}, \tag{5}$$

where D_{it} is a dummy variable indicating whether month t is the τ th month before or after (for negative and positive τ 's, respectively) the month that the policy was first introduced. For instance, $D_{it,\,0}$ equals 1 for the month of policy enactment for all policy-affected states and 0 otherwise. λ_τ serves to identify whether there is a similar pre-policy trend in both the policy-affected states and unaffected states and how the policy impacts change over time.

Specifically, we used $T = 12$ and $N = 12$ for Equation (5) to investigate the trends from 12 months before the policy (including the policy enactment month) and one year after the policy. To inspect the long-term trend of the policy, we also set a dummy variable, D_{after}, to represent 12 months after the policy enactment. Considering there might be an immediate impact of the policy, we treat relative time τ −3 to −1 before the treatment (i.e., one quarter before the policy) as the baseline to check the parallel trends.

As illustrated in Figure A2 of the Web Appendix, for the periods before the policy, all λ_τ ($\tau < 0$) that estimate the policy impacts on the overdose mortality of illicit drugs are largely around 0%, supporting the parallel trends. After the policy, the overdose mortality of illicit drugs dramatically increased compared to the pre-policy trend. Specifically, we could observe a sudden surge in the overdose mortality of illicit drugs in the month when the policy was enacted and two months after it ($\tau = 0$ and $\tau = 2$). The growth slightly dropped for the next two months but returned to high ratios (i.e., λ_τ) afterward until the end of the sample. Similar to Figure A2, Table A14 allows us to check the pre-treatment trends as well as the lead and lag effects of the RCL policy. The results suggest that pre-policy trends are unlikely to confound our policy impacts and bias our DID specification. In addition, we provide the relative time model at the quarterly level to better understand how the effects evolve over a longer term, as shown in Figure A3 and Table A15. The findings remain highly consistent.

To further confirm that the parallel trend assumption applies to other important dependent variables in this study, we also present the relative time model for the initiation and use of cannabis in Figure A4, the initiation and use of illicit drugs in Figure A5, and the Bitcoin transactions in Figure A6. Similar results are further exhibited in Tables A16 to A18 for cross-validation. Despite individual fluctuations, we consistently find that the parallel trend assumption holds across all these dependent variables, and the effects are significant after the RCL policy. This further supports our main effects and provides robustness checks to support our findings. In sum, these results suggest that the parallel trend assumption holds across these variables, and the policy indeed escalated the initiation and use of cannabis and illicit drugs as well as Bitcoin transactions by volume and value.

Alternative Specifications. Given most of the outcomes of interest are count numbers (e.g., number of overdose mortality), we leverage count models as an alternative specification to check the robustness of our results. To determine whether we should use a Poisson model or a Negative Binomial one, we look into the distribution of the number of overdose mortality and find that the mean is 45.91 while the variance is 1571.76, which shows a sign of overdispersion. We then run the test of the overdispersion parameter alpha and find that alpha is significantly different from zero, indicating that the Poisson distribution might not be appropriate and overdispersion possibly exists. Thus, we leverage Negative Binomial Regressions to replicate the primary results (in Tables 6 and 7). This approach also allows us to check the sensitivity of our results to a possible presence of overdispersion in our data (Lin et al., 2013).

Table A19 of the Web Appendix reports the policy impact on overdose mortality from illicit drugs, legal drugs, and alcohol. We find RCL increased illicit drug mortality, which is consistent with our previous results. We also find that RCL even has a negative effect on the mortality caused by legal drugs and alcohol, further ruling out the alternative explanation of the overall trend of substance abuse. In addition, this finding suggests that RCL failed to promote the substitution of illicit drugs for legal ones. Instead, our results suggest that RCL may have led to increased use of illicit drugs and even reduced use of legal drugs. Therefore, this result provides important support for the gateway effect. Table A20 in the Web Appendix reports the policy impact on the initiation and use of cannabis and illicit drugs, which are consistent with our main findings. In sum, the results of Negative Binomial Regressions are highly consistent with those in our main specifications.

There might be a concern that the difference-in-differences design with staggered treatment assignment may potentially lead to biased estimation. We leverage CSDID as an alternative specification to address this concern (e.g., Callaway and Sant'Anna, 2021). CSDID is an advanced approach to more accurately estimate the policy impacts implemented at different times, addressing potential estimation biases in traditional DID designs. As shown in Table A21 and Figure A7 in the Web Appendix, the results using CSDID remain largely consistent with our main analysis. In addition, although our treatment and control groups are strongly balanced, people may still be concerned about whether they are completely comparable. Thus, we leverage SDID (synthetic difference-in-differences) as an alternative method (Arkhangelsky et al., 2021). SDID combines synthetic controls with the DID design to create a more accurate control group, enabling effective causal inference even when the experimental and control groups are not perfectly comparable. As seen in Table A22 of the Web Appendix, the results using SDID consistently suggest that RCL led to a significant increase in overdose mortality caused by illicit drugs.

Alternative Drug Categorizations. Although the drug categorization in Table 3 has been used in previous studies (Powell et al., 2018; Team CPDO, 2013), some may still wonder if the categorization of illicit drugs is accurate since categories like T40.4 (Other synthetic narcotics) and T40.6 (other and unspecified narcotics) may include substances like fentanyl, which are legal for medical use but frequently associated with illicit drug use and overdoses. Thus, we re-categorize T40.4 and T40.6 as illicit drugs and replicate our analyses, as presented in Table A23 of the Web Appendix. The results do not take away our estimated impact of RCL and are consistent with our main findings using data categorizing T40.4 and T40.6 as non-illicit drugs.

Since our main analysis is based on drugs categorized in groups, another question is which specific drugs had increased or decreased after RCL, which requires analyzing the impact on a more granular categorization. Therefore, we conduct a by-subsect analysis by breaking down the illicit drugs by the level of harm: (1) the combination of T40.1 (heroin) and T40.5 (cocaine), the most hardcore illicit drugs, (2) the combination of T40.4 (other synthetic narcotics), and T40.6 (other and unspecified narcotics), which includes fentanyl and are legal for medical use but are frequently associated with illicit drug use and overdose. The results, as shown in Tables A24 and A25 of the Web Appendix, are consistent with our previous ones.

Controlling Google Search, Bitcoin Transactions, and Missing Values. There might be a concern that the RCL strongly overlaps with the Bitcoin adoption timeline, which may lead to biased estimation results. Although our DID design can rule out this concern since the Bitcoin adoption timeline is consistent between our treatment and control states, we further control for the Bitcoin adoption trend using the Google search popularity of "Bitcoin" as a proxy. Besides, people may wonder if there was a general interest in seeking illicit drugs on the darknet. To address this concern, we also utilize other keywords, including "darknet" and illicit drugs such as "cocaine" and "heroin." As shown in Table A26 of the Web Appendix, our results remain consistent with our main findings after controlling for all these keywords. The results also demonstrate the link between Bitcoin transactions and illicit drugs sold on darknets, especially heroin, which is consistent with the fact that heroin is the main cause of the second wave of the opioid overdose epidemic starting in 2010.[18]

We also control the Bitcoin transaction volume and value since it is a more direct way to control the general trend of Bitcoin adoption. The results, provided in Table A27 and Table A28 (using an alternative study period) of the Web Appendix, show consistent findings as our main analysis, as well as the analyses controlling the Google search keywords. In addition, since some states have no observations in one or several months, as listed in Table A29, there might be concerns that the missing values could alter the results. To this end, we conduct an additional test by filling the missing values with 0 and adding a dummy variable, which equals 1 if the observation is missing. This could help control the missing value issue. As shown in Table A30, the results are largely consistent with those in Table 8.

Leave-one-out Analysis. Since our main analysis estimates the average treatment effect of the RCL policy involving multiple RCL states, some may wonder if the results are driven by a single RCL state. To answer this question, we conduct a leave-one-out analysis, as shown in Table A31 of the Web Appendix. The results are largely consistent with our main findings and suggest that the impact of RCL is not sorely driven by a single state.

Extended Observation Window. Our observational window (2012.1–2018.12) was chosen to have consistent observational windows across all data sources used in this study. For example, NSDUH has no illicit drug initiation data earlier than 2012. However, some may want to see the results with earlier data for the pre-treatment trend as well as the most recent CDC WONDER data updated to the end of 2020. To this end, we extend our observational window to 2010–2020 (two years prior to and after our original observational window) for the analysis of the impact of RCL on the overdose mortality of illicit drugs. Particularly, we collect the most recent CDC WONDER data on overdose mortality of illicit drugs, which is updated to the end of 2020 in alignment with our new study period. The results are consistent with our main findings and are provided in Table A32 of the Web Appendix.

Impact of Unspecified Overdose Deaths. There might be a concern that many overdose death cases in the national data do not specify the substance involved, and this confound in the reporting of deaths may bias our results. To address this concern, we use the ICD-10 code R99 to obtain the number of deaths caused by unspecified causes on CDC WONDER. Then, we examine whether such deaths exhibit a significant difference between the treatment and control states. Our results shown in Table A33 of the Web Appendix suggest that such a difference is insignificant, suggesting that the number of deaths caused by unspecified causes would not confound our results.

Alternative Data Source of Drug Use. Since our NSDUH data are aggregated by state every two years for privacy protection purposes, people may be concerned that the evidence of increased use and initiation of illicit drugs based on NSDUH data is robust. To complement this analysis, we further collect annual data on admissions to substance use treatment facilities from the national Treatment Episode Data Set: Admissions (TEDS-A). We break down the treatment admissions by drug type (cannabis, cocaine, and heroin) and user type (first-time treatment and repeated treatment). Specifically, we use NOPRIOR in TEDS-A to classify admitted users by whether they have received prior treatment episodes in any substance use treatment program. We also use SUB1 in TEDS-A to denote the primary substance used among people admitted to substance use treatment facilities, including cannabis, cocaine, and heroin. The results by drug type and by user type results presented in Tables A34 and A35 of the Web Appendix yield consistent findings with those based on NSDUH data, suggesting that our findings on increased use and initiation of illicit drugs after RCL are robust.

5 Conclusion and Discussion

5.1 Key Findings and Implications

As the trend of legalizing recreational cannabis continues to spread in the United States, understanding its impact on public health becomes increasingly important to policymaking and social welfare. Despite its utmost importance, how RCL affects illicit drugs, being the substitution effect or the gateway effect, continues to attract ongoing heated debates without conclusive opinion and clear evidence. In this study, we address the void by examining the change in overdose mortality of illicit drugs after the policy. Our data contribute to the literature with much-needed empirical evidence to address the debate of RCL. Our findings confirm the gateway effect of RCL, which caused a significant increase in illicit drug-related deaths in affected states. More importantly, we investigate the reasonings behind the gateway effect and find supporting evidence that RCL encouraged users to initiate and use illicit drugs, increased purchases of illicit drugs on darknet markets, and led to higher risks of overdose mortality in affected states, especially those implemented a more lenient policy on recreational cannabis.

Our findings provide operation insights to legislators and policymakers. We first provide timely evidence that helps reconcile the public debate on RCL and suggest a valuable reference to states. For states that have not yet enacted RCL, in balancing seemingly economic benefits and potential harms to public health, we suggest these states evaluate whether the auxiliary resources are in place to control illicit drugs before legalizing recreational cannabis. Additionally, we identify the critical role of darknet markets in distributing illicit drugs. Bitcoin transactions traceable at the IP address level open a new arena for illicit drug control that can alleviate the potential harm of RCL to public health. We thus advocate spending more resources on recruiting cybersecurity experts and training law enforcement agencies to monitor, identify, and combat illicit drug transactions. Finally, it is worth noting that even though recreational cannabis became legalized in affected states, how lenient is the policy matters in determining the level of overdose mortality of illicit drugs. We recommend a strictly regulated dosage cap for recreational cannabis possession and a higher excise tax rate when implementing RCL, as our findings suggest a more lenient RCL policy would elevate the difficulty in controlling the overdose abuse of illicit drugs. These implications help policymakers pursue more prudent and informed decisions in RCL regulations.

5.2 Limitations and Future Research

Our study points several directions for future research. Policymaking involves high-stakes accountability and calls for evidence-based justifications. Different from an anecdotal discussion, empirical research using unique and innovative data sources such as Bitcoin transactions can bring much-needed insights to facilitate policymaking. Despite our earnest efforts in tracing Bitcoin transactions on darknet markets, not all these transactions might be associated with illicit drugs. Identifying Bitcoin transactions specifically related to illicit drugs is not a trivial task and has not been studied in the literature. It may require more fine-grained data for the purpose of model training and a more robust classification technique to deal with the higher data complexity. To obtain such data, researchers may dive into the Bitcoin blockchain as well as the darknet, and deanonymization approaches are also necessary because of the clandestine nature of these two contexts. Future studies exploring a classification strategy to identify illicit drug-related Bitcoin transactions will be very valuable in validating our findings further.

In addition, while this paper provides an innovative approach to utilize Bitcoin transactions to analyze the geolocations of illicit drug buyers, tracing the geolocation of Bitcoin transactions has become more challenging since Bitcoin changed from a gossip-style protocol to a diffusion spreading one in December 2015 (Fanti and Viswanath, 2017). Future research could explore how our method can be adapted to Bitcoin transactions beyond our study's observational period. As the drug overdose epidemic becomes more critical, we hope this paper will be a starting point, and our timely findings will stimulate more empirical research to inform policymaking in this growing area.

The public data used in this study also have some limitations. For example, it would be ideal if we could access a longer observational period of the use of illicit drug data. However, due to privacy reasons, NSDUH has imposed many restrictions on the disclosure of such data, and therefore, our analysis of the use of illicit drugs can only cover a relatively short period of time. We call for greater disclosure and sharing of such data while ensuring privacy. This will help stimulate research similar to this one and deepen the understanding of the true impact of RCL among policymakers and the public.

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Notes

1. In this study, our definition of illicit drugs includes primarily four main categories: heroin, cocaine, lysergide (LSD), and other unspecified hallucinogens. A more detailed definition can be found in Section 3.1.
2. In this study, our definition of legal drugs includes three categories of prescription opioids: natural opioid analgesics and semi-synthetic opioids, methadone, and other synthetic narcotics. A detailed definition is in Section 3.1.
3. Source: https://www.gminsights.com/industry-analysis/us-medical-marijuana-market, accessed on 1 May 2021.
4. Source: https://www.cdc.gov/marijuana/faqs.htm, accessed on 5 October 2021.
5. Source: https://itep.org/state-and-local-cannabis-tax-revenue-jumps-58-surpassing-3-billion-in-2020/, accessed on 28 June 2021.
6. A further discussion on this stream of literature supporting the substitution effect is provided in Table A2 of the Web Appendix. In particular, we draw a comparison between our finding (the gateway effect) and the contrasting findings in this stream of literature.
7. Source: https://www.unodc.org/res/WDR-2023/WDR23_B3_CH7_darkweb.pdf, accessed on 31 December 2023.
8. NSDUH is an annual nationwide survey on the use of legal and illegal drugs. It selects participants from all over the U.S. every year to estimate state-level survey outcomes about drug use in the population. To protect the confidentiality of respondents, the state-level estimates are only publicly disclosed on the Restricted-use Data Analysis System (RDAS) using a minimum of two years of pooled data. More details can be found at https://www.datafiles.samhsa.gov/get-help/codebooks/where-can-i-find-rdas-codebooks.
9. Despite there being 19 RCL-affected states in the U.S. as of September 2021 (the time of our analysis), the overdose mortality data of illicit drugs released by the CDC is only available for 2018 and earlier. For full availability of the data for analyses, we focused on the 11 states that are RCL-affected and have data on the overdose mortality of illicit drugs as of the end of 2018, yielding a study period between January 2012 and December 2018. Our Bitcoin transaction data spans from Dec 2014 to Dec 2015. See Section 3.3 for details. In addition, due to data source limitations, our NSDUH data only partially covers the study period from 2012 to 2018. See Table 2 for details.
10. T40 codes are a code family in the Tenth Revision of the International Classification of Disease (ICD-10), a medical classification list developed by the World Health Organization (WHO). T40 contains a group of drugs that alter a person's awareness and cause physical or psychological dependency (Team CPDO, 2013). A detailed explanation of how T40 and ICD-10 are used to access the data in WONDER is provided in Section A2 of the Web Appendix.
11. Each year, NSDUH randomly selects around 70,000 participants in all 50 states and the District of Columbia to answer a series of questions about drug use behaviors. For example, the respondents are asked whether they have ever used heroin and, if so, whether they used it in the current year or in the past two years (at the time the survey is conducted). Also, they are asked in what exact year and month did they first use heroin. Similar questions about using other drugs, such as cannabis and cocaine, are asked as well. The answers to these questions will be used to generate statistics and estimates at the national and state levels to measure drug use and health issues in the U.S., thereby helping the decision-making process of policies. For instance, NSDUH derives the numbers of respondents who used and initiated using heroin in the past year from their answers to heroin use questions. Then, the sample survey data is leveraged to generate state-level estimates of heroin use. The methodology and codebook can be found at https://www.datafiles.samhsa.gov/dataset/national-survey-drug-use-and-health-2019-nsduh-2019-ds0001.
12. We used cocaine to proxy the illicit drugs due to the unavailability of other illicit drug data such as RECHER2 (Past Year Initiation of Heroin Use). That is, the estimation of the impact of

RCL on the initiation of illicit drugs would be an underestimate. Illicit drug use data covers 2016, 2017, and 2018 since NSDUH did not report it until 2016; illicit drug initiation data covers 2013, 2015, 2016, 2017, and 2018 since NSDUH reported RECCOC2 bi-yearly before 2015.

13. Similar to using cocaine to represent illicit drugs, we use pain relievers, a type of prescription drug from doctors, to represent legal drugs. Other legal drug data, such as ABUPOSSTM (Prescription Stimulant Abuse in The Past Year), is not available. Both legal drug use and legal drug initiation data cover 2016, 2017, and 2018 since NSDUH did not report them until 2016.
14. In December 2015, Bitcoin switched to a diffusion protocol, where nodes independently relay information to all peers, and enhanced privacy by broadcasting simultaneously across the network, making it challenging to trace the original IP address of a transaction. However, it did not affect our data collection since most transactions collected from WalletExplorer.com in our study were in 2015 (Fanti and Viswanath, 2017).
15. Expect overdose mortality data from CDC WONDER and economic and demographic characteristics from ACS; the observations for other information are largely not available before 6 December 2012.
16. Due to data limitations, the unit of analysis of the NSDUH and TEDS-A data is state by year. The specific unit of analysis information is provided below each table and figure.
17. Calculation process: Given the average number of illicit drug overdose mortalities in treated states before the policy is 13.58, we use 13.58 to multiply 34.6%, resulting in 4.70 (13.58 × 34.6%) people per month per state. That is, the monthly illicit drug overdose mortality increases by about 34.6% or 4.7% people in an average affected state. Using 4.70 to multiple 12 months, we get an annual number of 56.4 people whose cause of mortality is illicit drug overdose in an average affected state.
18. Source: https://www.cdc.gov/opioids/data/analysis-resources.html, accessed on 1 November 2023.
19. In addition to log(illicit drug mortality$_{it}$) for 2012–2018, we have also expanded the study period to 2010–2020 for robustness check purposes. The summary statistics of log(illicit drug mortality$_{it}$) with the expanded study period are 2.15 (mean), 1.84 (std. dev.), 0.00 (min.), and 5.67 (max.).
20. T40.0 (opium) and T40.6 (other and unspecified narcotics) are classified as "Others" in the T40 family. T40.0 is usually the raw material of opioids rather than a produced drug. T40.6 is an opioid without information and cannot be classified as legal or illicit.
21. Information on all the RCL bills can be found at https://ballotpedia.org/Marijuana_on_the_ballotAll, accessed on 16 October 2021.

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